Balaji B

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.46
Petit Seminaire Higher Secondary School	Class XII (State Board)	2023	88.83%
Petit Seminaire Higher Secondary School	Class X (State Board)	2021	90.00%

SCHOLASTIC ACHIEVEMENTS

- Achieved the title of **Candidate Master** with a **peak rating of 1914** on the Codeforces platform.
- **Achieved AIR 159**, placing **6th within the college** among participating teams in **ICPC**.
- Secured **AIR 472** among nationwide participants in **IICPC QuantFest** and **qualified for the Regionals**.
- **Secured 2nd place** in **Dream11 Gameathon 1.0** among **5,000+ participants**; awarded **7.5 lakh**.
- **Finished 132nd** in the **Optiver Tradeathon,** a quantitative trading competition conducted among **IITs**.

PROJECTS

Netherlands Transit Router

- Built a **high-performance public transit routing engine** for the Netherlands using **live GTFS** data from **OVapi**, implementing the **McRAPTOR** multi-criteria **Pareto-optimal** routing algorithm.
- Developed a data pipeline to fetch, process, and serialize GTFS feeds into **compact binary formats**, covering **40 agencies**, **2,775 routes**, and **52,212 stops**, with **daily automated updates**.
- Architected the system end-to-end with a **C++ routing engine**, **FastAPI** backend, and **Leaflet.js** frontend for **interactive route visualization**.

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Custom eBPF CPU Scheduler (sched_ext)

- Built a **workload-specialized CPU scheduler** as an **eBPF** program using the Linux kernel's **sched_ext** framework, enabling **runtime scheduling policy changes** without kernel recompilation.
- Implemented the core scheduling pipeline (**enqueue**, **dispatch**, **select_cpu**) with **BPF maps** for per-task state, optimizing **time-slice allocation** and **CPU selection** for latency-sensitive workloads.
- Benchmarked against **EEVDF** using **schbench**, reducing **p99 tail latency by 20%** while maintaining throughput.
- **Prevented task starvation** and **priority inversion**, validated through **multi-workload stress tests**.

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Docker Clone (minicontainer)

- Built a lightweight container runtime from scratch in **C**, implementing process isolation using **PID**, **UTS**, **mount**, and **network namespaces** via clone()/unshare() system calls.
- **Enforced CPU and memory limits** using **cgroups v2**, replicating Docker's **--memory** and **--cpus** flags.
- Designed **OverlayFS-based copy-on-write images**, enabling **multiple containers** to share a **base image**.
- Added container networking via **veth pairs**, a **Linux bridge**, and **NAT** using **iptables** for outbound connectivity.

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Dream11 Fantasy Team Predictor

- Built a **fantasy cricket predictor** using **Random Forest Regression** on historical batting and bowling statistics.
- **Developed an end-to-end ML pipeline** for **feature engineering**, **model training**, and **Best XI generation**.
- Automated **squad detection**, player ranking, and **Captain/Vice-Captain** selection, exporting predictions to **CSV**.

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Sudoku Verifier

- Implemented a **multithreaded Sudoku verifier** in **C++** using **pthreads** and **locks** for concurrency control.
- **Parallelized** row, column, and subgrid validation, reducing **verification time** on **multicore systems**.
- Designed **thread-safe** error reporting and output generation, ensuring **consistent results** under concurrent execution.

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, Bash (Linux Shell Scripting), Verilog (HDL)

Developer Tools: Git, GitHub, Docker, Google Cloud Platform (GCP), Ubuntu Linux, LaTeX

Systems Programming: Linux Kernel, eBPF, POSIX Threads (pthreads), Namespaces, cgroups v2, OverlayFS

RELEVANT COURSES

Mathematics: Probability & Statistics, Linear Algebra, Graph Theory & Combinatorics, Machine Learning, Matrix Theory

Computer Science: Data Structures & Applications, Introduction to Programming, Operating Systems.

Electrical Engineering: Signal Processing, Control Systems, Digital Systems, Analog Devices

POSITIONS OF RESPONSIBILITY

Senior Core Member, Infero

Algorithmic and Logical Reasoning Club, IIT Hyderabad

- Curated and tested algorithmic problems for coding contests conducted by the club.
- Mentored junior students in competitive programming through training sessions and technical workshops.

OCS Coordinator, Office of Career Services, IIT Hyderabad

- Coordinated campus placement and internship drives for students.
- Managed student databases and supported placement-related operations.

EXTRACURRICULAR

- **Badminton:** Represented **IIT Hyderabad** at **Inter IIT** and competed at the **national level (U-15)**.
- **Guitar:** Passionate about playing acoustic guitar as a creative pursuit and stress-relieving hobby.
- Achieved a **2070** peak rating in Mental Math on **Matiks**.